

Subject Name: Bioinformatics lab

Subject No: BT39008

Prerequisite: Bioinformatics (BT31006)

Objective: To provide hands on knowledge on Bioinformatics algorithms to analyse sequence and structural data of macromolecules.

LTP- 0-0-3

Credit- 2

Content:

1. Understanding databases, understanding NCBI Entrez, Pairwise alignment and multiple alignment, BLAST analysis of nucleotide and protein, Genefinding, ORF finder, sequences

[3x3=9 hours]

2. Translation, secondary structure prediction, Data handling in PDB, Calculate structural and physicochemical parameters of macromolecules and their complexes

[3x3=9 hours]

3. Basis of molecular phylogeny, importance of 16S rRNA gene as molecular phylogeny marker, Phylogenetic tree construction and interpretations

[3x3=9 hours]

4. Understanding Next Generation Sequencing data, FASTQ data, Data Quality Check, Read Mapping, Analysis of Single Nucleotide polymorphisms from SAM files, RNA-seq data normalization, Differential Gene Expression analysis.

[3x3=9 hours]

Text books:

1. Bioinformatics: Sequence and Genome analysis David W. Mount
2. Introduction to Protein Structure, Carl Branden and John Tooze
3. Structural Bioinformatics by Jenny Gu and Philip E. Bourne
4. Bioinformatics and Functional Genomics by J. Pevsner
5. High-Throughput Next Generation Sequencing, Methods and Applications. (Springer). Editors: Kwon, Young Min, Ricke, Steven C. (Eds.)